Scaling Laws at the Edge: First-Principles Adaptation, Subspace Allocation, and Quantization Trade-offs for Small Language Models

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Abstract

Adapting and deploying Small Language Models (SLMs, 26M–1.5B parameters) on memory- and compute-constrained edge accelerators is commonly treated as a scaled-down version of frontier LLM fine-tuning. This scaling assumption is not innocent: in the sub-billion regime, unconstrained full-parameter updates overfit compact domain corpora, attention-only adapters under-parameterize factual recall, and mixed-precision quantization kernels introduce synchronization latencies that outweigh memory-bandwidth savings. We formulate a first-principles framework grounded in the hardware Roofline model, sample-limited intrinsic rank bounds, Transformer subspace specialization, and preference trust-region dynamics. Across a controlled 39-run empirical study spanning four model scales (`MiniMind2-Small-26M`, `MiniMind2-104M`, `Qwen2.5-0.5B`, and `Qwen2.5-1.5B`) on an NVIDIA Turing (`sm_75`) GPU across three zero-leakage diagnostic tasks—multi-hop domain QA (CourseQA), schema-constrained extraction (JSONOut), and out-of-template classification (Intent)—we establish four principal results. First, low-rank adaptation acts as an implicit spectral regularizer: on `MiniMind-26M`, LoRA ($r = 16$) improves CourseQA score from 36.80% (8-epoch Full SFT) to 45.71% (+8.91%) while reducing peak training memory by 65.1%, and rank ablation on `Qwen2.5-0.5B` peaks at $r = 4$ (56.88%). Second, subspace allocation exhibits a sharp task dichotomy: query-value (`qv`) adapters saturate syntactic schema extraction (100.0%), whereas factual QA improves monotonically from `qv` (18.87%) to `qkvo` (27.17%) and full linear including SwiGLU MLP projections (32.23%). Third, batch-1 decoding on Turing Tensor Cores is governed by dequantization kernel overhead rather than footprint alone: 4-bit NormalFloat (`NF4`) saves 505.6 MB of static memory while achieving 2.77× higher decoding throughput (16.01 tok/s) than outlier-splitting `LLM.int8()` (5.77 tok/s). Fourth, Direct Preference Optimization (DPO) on compact models undergoes an acute step-size phase transition, where bounding the learning rate from 5×10^{-5} to 5×10^{-6} eliminates off-manifold likelihood displacement and doubles classification accuracy over the zero-shot baseline.

1 Introduction

Frontier Large Language Models (LLMs) achieve strong zero-shot generalization across open-domain tasks (Brown et al., 2020; Achiam et al., 2023), but their multi-gigabyte memory footprints and high time-to-first-token (TTFT) preclude deployment under strict hardware budgets. In edge workstations, industrial controllers, and single-GPU cloud instances, Small Language Models (SLMs) spanning 10^7 to 1.5×10^9 parameters (Yang et al., 2024; Gong, 2025) offer a viable alternative when adapted to well-scoped downstream tasks such as technical question answering, schema-constrained JSON telemetry extraction, and deterministic intent routing.

However, directly transferring fine-tuning and post-training quantization recipes from 7B–70B models to the sub-billion regime frequently produces counterintuitive regressions. When operating under tight parameter capacities and first-generation Tensor Core microarchitectures (e.g., NVIDIA Turing `sm_75`), four structural tensions emerge:

- **Over-parameterized updates vs. sample complexity.** Because a 26M-parameter model fits in < 500 MB of training VRAM, full-parameter supervised fine-tuning (Full SFT) is computationally inexpensive. Yet when the downstream corpus contains $N \sim 10^2$ domain examples, unconstrained rank-d_{model} gradient updates memorize surface phrasing in trailing singular directions rather than isolating the low-dimensional task signal (Aghajanyan et al., 2021; Hu et al., 2022).
- **Syntactic routing vs. factual memory subspaces.** Standard parameter-efficient fine-tuning (PEFT) conventions attach low-rank adapters only to attention query and value projections (`qv`) (Hu et al., 2022). While sufficient for re-routing surface syntax, this leaves feed-forward network (FFN) key-value memories (Geva et al., 2021; Meng et al., 2022) frozen during domain knowledge injection.
- **Memory bandwidth vs. dequantization kernel overhead.** Weight-only quantization (`INT8`, `NF4`) reduces static VRAM (Dettmers et al., 2022, 2023). At batch size $B = 1$, however, the wall-clock time saved by transferring fewer bytes across memory bus B_{mem} on a 0.5B model is only 1–2 ms/token (Williams et al., 2009); any dynamic

activation-thresholding or host-device synchronization overhead directly inflates decoding latency.

- **Off-manifold likelihood displacement in preference alignment.** Direct Preference Optimization (DPO) (Rafailov et al., 2023) eliminates the explicit reward model by contrasting chosen and rejected completions directly in policy log-space. On compact models with shared prompt prefixes, aggressive negative gradients rapidly drain probability mass from valid label tokens toward out-of-vocabulary strings (Pal et al., 2024; Tajwar et al., 2024).

Problem statement and system model. Given a pre-trained causal Transformer π_{θ_0} with parameter count $\Phi \in [2.6 \times 10^7, 1.54 \times 10^9]$, a target task distribution $\mathcal{D}$ with N labeled training examples, and a hardware accelerator characterized by peak compute C_{peak}, memory bandwidth B_{mem}, and physical memory ceiling $M_{\max}$, we seek an adaptation operator $\theta_0 \mapsto \theta = \theta_0 + \Delta\theta$ and an inference quantization bit-width $b \in \{4, 8, 16\}$ that maximize out-of-sample task performance $\mathcal{P}(\pi_\theta)$ subject to $M_{\text{peak}}(\Phi, b) \leq M_{\max}$ and per-token latency constraints. Figure 1 summarizes our analytical framework.

Contributions.

- **First-principles latency and rank bounds for SLMs.** We derive the exact dequantization latency inequality under the Roofline model (Proposition 1) and the sample-limited numerical rank bound for assistant-masked updates (Proposition 2), explaining why low-rank constraints act as a spectral regularizer in compact regimes (§2).
- **Task-dependent subspace hierarchy.** We formalize and empirically verify a functional dichotomy between attention routing ($\mathcal{M}_{\text{attn}}$) and FFN key-value storage ($\mathcal{M}_{\text{ffn}}$): `qv` adapters saturate strict JSON schema extraction (100.0%), whereas domain QA requires adapting FFN projections for a +13.36% gain over `qv` (§2.3, §4.3).
- **Controlled 39-run scaling, quantization, and alignment audit.** Across four model scales (26M to 1.5B) and three zero-leakage diagnostic tasks, we map the latency–memory Pareto frontier of `FP16`, `LLM.int8()`, and `NF4`, and identify the exact step-size phase transition ($\eta = 5\times10^{-5} \to 5\times10^{-6}$) that prevents likelihood displacement in compact DPO (§3).

2 First-Principles Theoretical Framework

2.1 Hardware Roofline and Memory State Decomposition

Let Φ denote the total parameter count of a causal Transformer deployed on an accelerator with peak half-precision compute C_{peak} and memory bandwidth B_{mem}. On an NVIDIA Tesla T4 (`sm_75`, 40 SMs, 14.56 GiB GDDR6), $C_{\text{peak}}^{\text{FP16}} = 65$ TFLOPS and $B_{\text{mem}} = 320$ GB/s, placing the Roofline ridge point at $\mathcal{I}^* = C_{\text{peak}}/B_{\text{mem}} = 203.1$ FLOPs/byte (Williams et al., 2009). Because Turing Tensor Cores lack native hardware `bfloat16` pipelines and Ampere-only `sm_80+` instructions required by FlashAttention-2 (Dao, 2024), hardware-optimal execution requires `float16` weights paired with memory-efficient Scaled Dot-Product Attention (SDPA) (Rabe and Staats, 2021).

Training memory decomposition. Full-parameter mixed-precision SFT with AdamW (Loshchilov and Hutter, 2019) requires storing `FP16` parameters (2Φ bytes), `FP16` gradients (2Φ bytes), and `FP32` master weights plus first and second moments (m_t, v_t, totaling 12Φ bytes):

$$M_{\text{train}}^{\text{Full}}(\Phi) = 16\Phi + M_{\text{act}}(B, L, d_{\text{model}}). \tag{1}$$

For $\Phi = 2.6 \times 10^7$ (`MiniMind-26M`), $16\Phi \approx 416$ MB, enabling full-parameter training in < 0.5 GB VRAM. Conversely, for $\Phi = 1.54 \times 10^9$ (`Qwen2.5-1.5B`), $16\Phi \approx 24.6$ GB, exceeding the 14.56 GiB physical limit prior to activation allocation. Freezing the base model in 4-bit NormalFloat with double quantization ($b \approx 0.55$ bytes/param) and updating only low-rank adapters $\Phi_{\text{LoRA}} \ll \Phi$ (Dettmers et al., 2023) reduces training state memory to:

$$M_{\text{train}}^{\text{QLoRA}}(\Phi) = 0.55\Phi + 16\Phi_{\text{LoRA}} + M_{\text{act}}^{\text{ckpt}}, \tag{2}$$

bounding the static and optimizer state footprint of a 1.54B model below 1.2 GB.

Proposition 1 (Quantized Decoding Latency Condition)**.** *During batch-1 autoregressive decoding ($B = 1$), generating each token requires 2Φ FLOPs and streams $b\Phi$ bytes of model weights from global memory, resulting in arithmetic intensity $\mathcal{I}_{decode} = 2/b \ll \mathcal{I}^*$. Let $t_{dequant}(b)$ denote the per-token kernel execution and synchronization overhead of dequantizing b-bit weights to `FP16`. Weight quantization from $b_0 = 2$ (`FP16`) to $b < 2$ reduces per-token decoding latency $t_{token}(b)$ if and only if:*

$$t_{dequant}(b) < \Delta t_{mem}(b) \triangleq \frac{(2-b)\Phi}{B_{mem}}. \tag{3}$$

Derivation. Under the memory-bound regime of the Roofline model ($\mathcal{I}_{\text{decode}} \leq 4 \ll 203.1$), ideal memory-transfer time is $b\Phi/B_{\text{mem}}$. Adding dequantization kernel time $t_{\text{dequant}}(b)$ yields $t_{\text{token}}(b) \approx b\Phi/B_{\text{mem}} + t_{\text{dequant}}(b) + t_{\text{compute}}$. Comparing $t_{\text{token}}(b) < t_{\text{token}}(2)$ with $t_{\text{dequant}}(2) = 0$ immediately yields Inequality (3). □

For a sub-billion model ($\Phi = 4.94 \times 10^8$, `Qwen2.5-0.5B`) on T4 ($B_{\text{mem}} = 320$ GB/s), reducing weight precision from `FP16` ($b = 2$) to `INT8` ($b \approx 1.1$) or `NF4` ($b \approx 0.55$) yields a theoretical memory-transfer

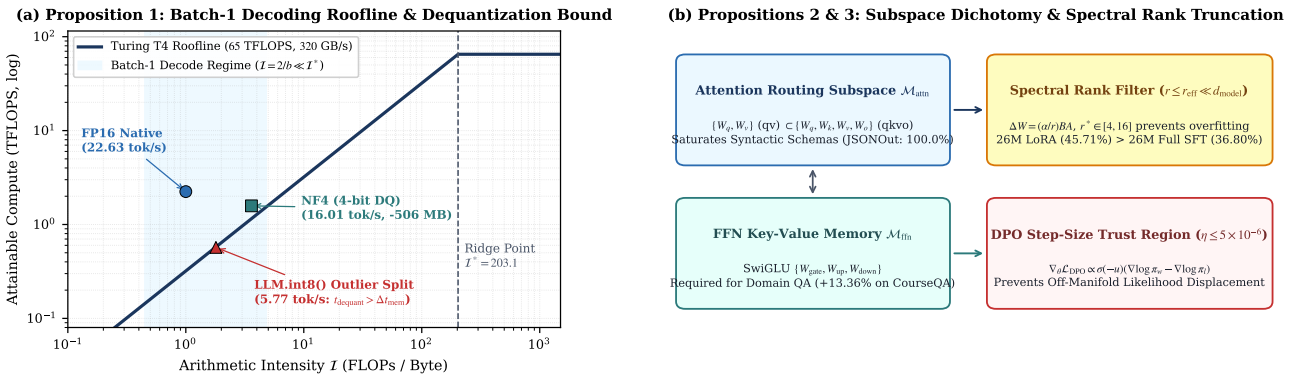


Figure 1: First-principles adaptation and deployment framework for edge SLMs. (a) Hardware Roofline model for NVIDIA Turing `sm_75` (65 TFLOPS, 320 GB/s). Batch-1 decoding operates in the memory-bound regime ($\mathcal{I} \ll \mathcal{I}^*$), where `LLM.int8()` outlier decomposition incurs synchronization overhead exceeding bandwidth savings (Proposition 1). **(b)** Structural decomposition of Transformer adaptation into attention routing ($\mathcal{M}_{\text{attn}}$), FFN key-value memory ($\mathcal{M}_{\text{ffn}}$), spectral rank truncation ($r \leq r_{\text{eff}}$), and DPO trust-region step control.

reduction Δt_{mem} of only 1.39 ms to 2.24 ms per token. Any quantization scheme that incurs dynamic per-layer outlier extraction and host-device stream synchronization (`LLM.int8()`; Dettmers et al., 2022) with $t_{\text{dequant}} \gg 2.2$ ms will strictly slow down inference despite reducing VRAM.

2.2 Intrinsic Rank Bound and Spectral Regularization

For a pre-trained projection $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, LoRA (Hu et al., 2022) constrains the parameter update to $\Delta W = (\alpha/r)BA$, where $B \in \mathbb{R}^{d_{\text{out}} \times r}$, $A \in \mathbb{R}^{r \times d_{\text{in}}}$, and $r \ll \min(d_{\text{in}}, d_{\text{out}})$.

Proposition 2 (Sample-Limited Rank Upper Bound)**.** *Consider assistant-masked cross-entropy training over N examples with average target completion length $\bar{L}_{ans}$ tokens. At each layer, the cumulative first-order gradient update $\Delta W_{SFT} = -\eta \sum_{i=1}^{N} \sum_{t=1}^{\bar{L}_{ans}} g_{i,t} h_{i,t}^\top$ is a linear combination of at most $N\bar{L}_{ans}$ rank-1 outer products between backpropagated output gradients $g_{i,t} \in \mathbb{R}^{d_{out}}$ and input activations $h_{i,t} \in \mathbb{R}^{d_{in}}$. Its numerical rank at singular-value threshold $\epsilon > 0$ satisfies:*

$$r_{eff}(\Delta W_{SFT}, \epsilon) \leq \min\big(d_{in},\, d_{out},\, \kappa(\epsilon) N \bar{L}_{ans}\big), \quad (4)$$

where $\kappa(\epsilon) \in (0, 1]$ is the fraction of linearly independent token-level gradient directions exceeding ϵ.

When adapting an SLM on a compact domain dataset ($N \approx 10^2$, $\bar{L}_{\text{ans}} \approx 25$), strong syntactic and semantic correlation across examples concentrates the task-relevant gradient energy into the top $r^* \in [4, 16]$ singular vectors ($\kappa(\epsilon) \ll 0.01$). Unconstrained Full SFT updates all $\min(d_{\text{in}}, d_{\text{out}}) \geq 512$ singular directions, allowing trailing singular vectors to memorize idiosyncratic lexical artifacts of the training split. Enforcing $\text{rank}(\Delta W) \leq r$ acts as a hard spectral truncation filter that projects updates onto the principal task subspace while leaving the orthogonal complement $\ker(A)$ invariant.

2.3 Subspace Specialization: Attention vs. FFN

Each Transformer layer contains attention projections $\mathcal{M}_{\text{attn}} = \{W_q, W_k, W_v, W_o\}$ and gated MLP/FFN projections $\mathcal{M}_{\text{ffn}} = \{W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}\}$. Mechanistic interpretability studies (Geva et al., 2021; Meng et al., 2022) show that FFN layers act as key-value memories where $W_{\text{gate}}, W_{\text{up}}$ detect input patterns and W_{down} projects factual value vectors into the residual stream, whereas attention heads route contextual tokens and govern output syntax. This structural division implies two distinct adaptation regimes:

1. **Syntax-constrained extraction (JSONOut, Intent).** All entity values are explicitly supplied in the prompt context; adaptation only needs to suppress conversational preamble and enforce output schema grammar. Adapting $\mathcal{M}_{\text{qv}} = \{W_q, W_v\}$ suffices to saturate test accuracy: $\mathcal{P}(\mathcal{M}_{\text{qv}}) \approx \mathcal{P}(\mathcal{M}_{\text{attn}} \cup \mathcal{M}_{\text{ffn}})$.
2. **Knowledge-intensive QA (CourseQA).** Answering multi-hop technical questions without external context requires modifying factual associations stored in FFN weights, inducing a strict monotonic ordering:

$$\mathcal{P}(\mathcal{M}_{\text{qv}}) < \mathcal{P}(\mathcal{M}_{\text{attn}}) < \mathcal{P}(\mathcal{M}_{\text{attn}} \cup \mathcal{M}_{\text{ffn}}). \quad (5)$$

2.4 Likelihood Displacement in Compact DPO

Given preference triples (x, y_w, y_l), DPO (Rafailov et al., 2023) optimizes:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}\Bigg[\log \sigma\Big(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\Big)\Bigg]. \quad (6)$$

Dataset	Task Regime	Train	Dev	Test	Total
CourseQA	Multi-hop domain QA (8 topics)	104	34	32	170
JSONOut	Schema extraction (4 schemas)	384	128	128	640
Intent	Out-of-template routing (6 classes)	576	192	192	960
Total	**Zero-leakage benchmark**	**1,064**	**354**	**352**	**1,770**

Table 1: Diagnostic benchmark statistics. Cross-split ID overlap is strictly $\emptyset$ across all tasks; Intent test queries use held-out syntactic templates disjoint from train and dev.

Let $u(x, y_w, y_l) = \beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}$. The gradient with respect to θ is:

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\beta\, \mathbb{E}\Big[\sigma(-u)\big(\nabla_\theta \log \pi_\theta(y_w|x) - \nabla_\theta \log \pi_\theta(y_l|x)\big)\Big]. \tag{7}$$

In discrete classification (Intent), y_w and y_l are short snake_case label tokens conditioned on an identical prompt x, so their hidden-state representations $\nabla_\theta \log \pi_\theta(y_w|x)$ and $\nabla_\theta \log \pi_\theta(y_l|x)$ exhibit high cosine similarity in the final projection layer. When the learning rate η exceeds the local trust region ($\eta = 5 \times 10^{-5}$), the repulsive gradient $-\nabla_\theta \log \pi_\theta(y_l|x)$ destabilizes shared token embeddings, shifting probability mass away from the entire valid label set $\mathcal{Y}$ toward arbitrary out-of-vocabulary tokens (likelihood displacement; Pal et al., 2024; Tajwar et al., 2024). Preventing collapse without an explicit SFT stage requires constraining $\eta \leq 5 \times 10^{-6}$.

3 Experimental Setup

We organize our empirical evaluation around four falsifiable questions corresponding to Section 2: **Q1 (Spectral regularization)**: Does rank-restricted LoRA outperform unconstrained Full SFT on compact QA corpora, and how does optimal rank scale? **Q2 (Subspace dichotomy)**: When do attention-only (`qv`) adapters suffice, and when must FFN projections (`all-linear`) be unlocked? **Q3 (Quantized decoding Roofline)**: Does empirical batch-1 throughput on Turing Tensor Cores obey Proposition 1 across `FP16`, `LLM.int8()`, and `NF4`? **Q4 (DPO trust-region stability)**: How does learning-rate scaling govern likelihood displacement vs. boundary alignment in compact DPO?

3.1 Zero-Leakage Diagnostic Benchmark

To rule out pre-training memorization and evaluate genuine task adaptation, we construct a 1,770-sample benchmark (Table 1) with deterministic seed 42 and programmatic verification that train/dev/test ID intersections are strictly empty:

- **CourseQA ($N = 170$).** Technical question-answer pairs stratified across eight computer science domains (data structures, asymptotic complexity, operating systems, computer networks, relational databases, boolean logic, sorting, and graph algorithms). Each item is evaluated via normalized Exact Match (EM), character-level Token-F1, and answer containment (Contains). To reflect question structure accurately, each item carries a primary scorer tag $\text{scorer}(i) \in \{\text{EM}, \text{F1}\}$ (EM for short factoid/complexity items; Token-F1 for multi-clause definitional items), and the headline Overall score is the exact per-sample average $\frac{1}{|D_{\text{test}}|} \sum_i s_{\text{scorer}(i)}(\hat{y}_i, y_i)$.
- **JSONOut ($N = 640$).** Unstructured natural language descriptions mapped to four strict JSON schemas (`course_info`, `student_grade`, `lab_schedule`, `bug_report`) requiring semantic-to-boolean type conversion (e.g., mapping "pending fix" to `resolved: false`). We report JSON Parse Rate (`json.loads` success), strict Schema Valid Rate (exact key set and Python types, used as Overall), and key-level Field Accuracy.
- **Intent ($N = 960$).** Utterances balanced across six routing classes (`deadline_extension`, `grade_appeal`, `concept_question`, `debug_help`, `env_setup`, `team_matching`). While train and dev splits are generated from declarative templates, the 192 test queries use disjoint interrogative and colloquial templates. We report Accuracy (Overall), Macro-F1, and out-of-vocabulary Unmapped Rate.

3.2 Models, Masking, and Compute Protocol

We evaluate four open-weight checkpoints spanning 59× in parameter scale: `MiniMind2-Small` (26M), `MiniMind2` (104M) (Gong, 2025), `Qwen2.5-0.5B-Instruct` (494M), and `Qwen2.5-1.5B-Instruct` (1.54B) (Yang et al., 2024). All 39 runs execute on a single NVIDIA Tesla T4 GPU (14.56 GiB GDDR6, Turing `sm_75`) under Python 3.13.16, PyTorch 2.11.0+cu130, Transformers 4.57.6, and PEFT 0.21.2.

To prevent prompt-token reconstruction from dominating gradients on short target completions, our collator enforces exact assistant-only loss masking ($y_t = -100$ on all system and user tokens prior to `<|im_start|>assistant\n`). In mixed-precision LoRA/QLoRA runs, all trainable adapter weights are cast to `float32` master weights to maintain `GradScaler` numerical stability. Evaluation uses greedy decoding (`do_sample=False`) with peak VRAM tracked via `torch.cuda.max_memory_allocated()`.

4 Main Results and Analysis

4.1 Scaling Ladder and Spectral Regularization (Q1)

Table 2 and Figure 2(a) report performance across all four model tiers. Three findings stand out:

Model Scale	Task	Method	Schedule / Rank	Primary Metric 1	Primary Metric 2	Overall	Gain (Δ)	Time (s)	Peak VRAM (MB)
Tier 1: Ultra-Compact Models (`MiniMind2-Small`, $\Phi = 26M$ & `MiniMind2`, $\Phi = 104M$)									
`MiniMind-26M`	CourseQA	Zero-Shot	—	EM: 0.0000	F1: 0.1260	0.1025	—	80.03	60.0
`MiniMind-26M`	CourseQA	Full SFT	3 ep / Full	EM: 0.0625	F1: 0.2123	0.1870	+0.0845	4.37	473.7
`MiniMind-26M`	CourseQA	Full SFT	8 ep / Full	EM: 0.2812	F1: 0.4055	0.3680	+0.2655	5.57	375.0
`MiniMind-26M`	CourseQA	**LoRA**	8 ep / $r = 16$	**EM: 0.4062**	**F1: 0.5370**	**0.4571**	**+0.3546**	16.22	**130.9**
`MiniMind-104M`	CourseQA	Zero-Shot	—	EM: 0.0000	F1: 0.1458	0.1034	—	29.09	531.7
`MiniMind-104M`	CourseQA	Full SFT	8 ep / Full	EM: 0.3438	F1: 0.4663	0.4256	+0.3222	14.00	1431.5
`MiniMind-26M`	JSONOut	Zero-Shot	—	Parse: 0.0391	Field: 0.0016	0.0000	—	253.32	133.1
`MiniMind-26M`	JSONOut	**Full SFT**	4 ep / Full	**Parse: 1.0000**	**Field: 1.0000**	**1.0000**	**+1.0000**	105.13	372.1
`MiniMind-26M`	Intent	Zero-Shot	—	Acc: 0.0000	F1: 0.0000	0.0000	—	46.00	370.9
`MiniMind-26M`	Intent	**Full SFT**	4 ep / Full	**Acc: 1.0000**	**F1: 1.0000**	**1.0000**	**+1.0000**	11.80	376.0
Tier 2: Sub-Billion & Billion-Scale Models (`Qwen2.5-0.5B-Instruct`, $\Phi = 494M$ & `Qwen2.5-1.5B-Instruct`, $\Phi = 1.54B$)									
`Qwen2.5-0.5B`	CourseQA	Zero-Shot	—	EM: 0.0000	F1: 0.1662	0.1178	—	42.38	1012.8
`Qwen2.5-0.5B`	CourseQA	**LoRA**	6 ep / $r = 16$	**EM: 0.5000**	**F1: 0.6467**	**0.5619**	**+0.4441**	15.14	2032.8
`Qwen2.5-0.5B`	CourseQA	QLoRA (NF4)	6 ep / $r = 16$	EM: 0.4375	F1: 0.6148	0.5167	+0.3989	15.24	**1771.8**
`Qwen2.5-1.5B`	CourseQA	Zero-Shot (NF4)	—	EM: 0.0000	F1: 0.1972	0.1425	—	58.24	2176.0
`Qwen2.5-1.5B`	CourseQA	**QLoRA (NF4)**	5 ep / $r = 16$	**EM: 0.5000**	**F1: 0.7057**	**0.5880**	**+0.4455**	18.79	4715.1
`Qwen2.5-0.5B`	JSONOut	Zero-Shot	—	Parse: 1.0000	Field: 0.8391	0.5703	—	290.62	1774.1
`Qwen2.5-0.5B`	JSONOut	**LoRA**	4 ep / $r = 16$	**Parse: 1.0000**	**Field: 1.0000**	**1.0000**	**+0.4297**	231.46	2026.2
`Qwen2.5-0.5B`	Intent	Zero-Shot	—	Acc: 0.2188	F1: 0.1903	0.2188	—	25.57	2024.8
`Qwen2.5-0.5B`	Intent	**LoRA**	4 ep / $r = 16$	**Acc: 1.0000**	**F1: 1.0000**	**1.0000**	**+0.7812**	26.02	2025.4

Table 2: Main results across the four-tier SLM ladder on held-out test splits (CourseQA $n = 32$, JSONOut $n = 128$, Intent $n = 192$). For JSONOut, Overall is strict Schema Valid Rate; for CourseQA, Overall is the question-type-dispatched EM/F1 score. Best adapted results per model tier and task are bolded.

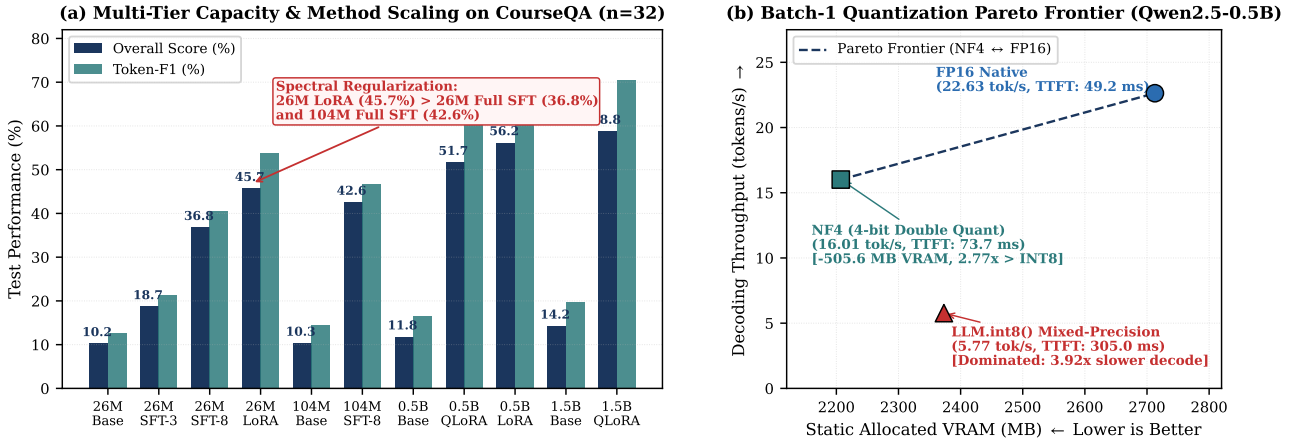


Figure 2: Capacity scaling and quantization Pareto frontier. (a) On CourseQA, rank-16 LoRA on `MiniMind-26M` (45.71%) outperforms both 8-epoch Full SFT on `26M` (36.80%) and 8-epoch Full SFT on `104M` (42.56%), confirming low-rank spectral regularization (Proposition 2). **(b)** Batch-1 quantization benchmark on `Qwen2.5-0.5B`: `NF4` and `FP16` form the latency–memory Pareto frontier, whereas `LLM.int8()` is strictly dominated due to outlier-decomposition synchronization overhead (Proposition 1).

Low-rank adaptation beats Full SFT on compact domain corpora. On `MiniMind-26M` (CourseQA), increasing Full SFT from 3 epochs to 8 epochs raises Overall from 18.70% to 36.80% (EM 28.12%, Token-F1 40.55%). Under the exact same 8-epoch budget, restricting updates to a rank-16 LoRA subspace achieves 45.71% Overall (EM 40.62%, Token-F1 53.70%, Contains 43.75%)—an +8.91% absolute improvement over 8-epoch Full SFT on the same 26M backbone, and +3.15% higher than 8-epoch Full SFT on the 4× larger `MiniMind-104M` model (42.56%). Simultaneously, LoRA reduces peak VRAM from 375.0 MB to 130.9 MB (−65.1%). Consistent with Proposition 2, rank truncation prevents small-sample memorization while preserving pre-trained representations.

Sub-billion SLMs achieve exact mastery on constrained syntax and out-of-template routing. Zero-shot baselines struggle with strict output constraints: `MiniMind-26M` scores 0.00% schema validity on JSONOut (3.91% parse rate) and 0.00% accuracy on Intent (100.0% unmapped conversational replies). `Qwen2.5-0.5B` parses valid JSON (100.0%) but achieves only 57.03% strict schema validity due to boolean type violations (emitting raw Chinese status tokens instead of JSON booleans), and scores 21.88% on Intent (38.54% unmapped rate). After 4 epochs of assistant-masked fine-tuning, both `MiniMind-26M` (Full SFT) and `Qwen2.5-0.5B` (LoRA) reach 100.0% JSON parse rate, 100.0% strict schema validity, 100.0% field accuracy, and 100.0% out-of-template intent classification accuracy. Moreover, suppressing verbose zero-shot continuations reduces `MiniMind-26M` evaluation latency on Intent from 46.00 s to 11.80 s (3.9× speedup).

Monotonic capacity scaling on knowledge-intensive QA. On CourseQA, adapted performance scales monotonically with pre-trained capacity: from 45.71% (`MiniMind-26M` LoRA) to 56.19% (`Qwen2.5-0.5B` LoRA, EM 50.00%, Token-F1 64.67%)

Variant	Static MB	Peak MB	TTFT (ms)	Prefill/s	Decode/s	QA Score
FP16 (Native)	2712.7	2716.4	**49.21**	**835.64**	**22.63**	0.1178
LLM.int8()	2373.1	2378.8	305.02	190.33	5.77	**0.1224**
NF4 (4-bit DQ)	**2207.1**	**2217.3**	73.65	578.19	16.01	0.1105

Table 3: Batch-1 inference quantization benchmark on `Qwen2.5-0.5B` (CourseQA test split, Tesla T4). Static MB and Peak MB denote allocated CUDA memory before and during generation.

Ablation Axis	Configuration	Metric 1	Metric 2	Overall	VRAM (MB)
(A) Training Sample Size N (MiniMind-26M, CourseQA, 8 ep Full SFT)					
Sample Size	$N = 25$ (24%)	EM: 0.0625	F1: 0.0931	0.0767	375.0
Sample Size	$N = 50$ (48%)	EM: 0.2188	F1: 0.2894	0.2675	376.9
Sample Size	$N = 104$ (100%)	**EM: 0.2812**	**F1: 0.4055**	**0.3680**	375.0
(B) Training Label Corruption η_{noise} (MiniMind-26M, Intent, 4 ep Full SFT)					
Label Noise	$\eta_{\text{noise}} = 0\%$	**Acc: 1.0000**	**F1: 1.0000**	**1.0000**	376.8
Label Noise	$\eta_{\text{noise}} = 20\%$	Acc: 0.9792	F1: 0.9783	0.9792	375.6
Label Noise	$\eta_{\text{noise}} = 50\%$	Acc: 0.5677	F1: 0.4743	0.5677	376.0
(C) LoRA Rank r (Qwen2.5-0.5B, CourseQA, 6 ep, $\alpha = 2r$, all-linear)					
Adapter Rank	$r = 4, \alpha = 8$	**EM: 0.5000**	**F1: 0.6995**	**0.5688**	**1948.1**
Adapter Rank	$r = 8, \alpha = 16$	EM: 0.4688	F1: 0.6525	0.5221	1972.0
Adapter Rank	$r = 16, \alpha = 32$	EM: 0.4688	F1: 0.6664	0.5257	2022.6
Adapter Rank	$r = 32, \alpha = 64$	EM: 0.4375	F1: 0.6243	0.5148	2122.9

Table 4: Orthogonal ablations on sample size (A), label corruption (B), and LoRA rank (C). All runs fix random seed 42.

and 58.80% (`Qwen2.5-1.5B` QLoRA, EM 50.00%, Token-F1 70.57%). Using 4-bit `NF4` double quantization (Eq. 2), `Qwen2.5-1.5B` QLoRA training peaks at 4,715.1 MB VRAM (31.6% of T4 capacity), and on `Qwen2.5-0.5B`, QLoRA saves 261.0 MB peak training VRAM over `FP16` LoRA (1,771.8 MB vs. 2,032.8 MB) while retaining 92.0% of its relative QA gain.

4.2 Quantized Decoding Roofline and Pareto Frontier (Q3)

Table 3 and Figure 2(b) compare native `FP16`, 8-bit mixed-precision decomposition (`LLM.int8()`; Dettmers et al., 2022), and 4-bit NormalFloat with double quantization (`NF4`; Dettmers et al., 2023) on `Qwen2.5-0.5B`.

The measurements directly verify Proposition 1. Although `LLM.int8()` reduces static memory by 339.6 MB (2,712.7 → 2,373.1 MB) and preserves zero-shot QA accuracy (0.1224), it inflates TTFT by 6.20× (305.02 ms vs. 49.21 ms) and reduces decoding throughput by 3.92× (5.77 tok/s vs. 22.63 tok/s). Because `LLM.int8()` inspects activation magnitudes at every linear layer to route outlier channels ($|\mathbf{x}_j| > 6.0$) into `FP16` GEMMs before merging with `INT8` outputs, the per-token synchronization penalty ($t_{\text{dequant}} \approx$ 129.1 ms/tok) overwhelms the ∼ 1.4 ms/tok GDDR6 bandwidth savings on a 494M model.

In contrast, 4-bit `NF4` relies on uniform blockwise lookup-table dequantization without dynamic activation branching. `NF4` achieves the lowest static memory footprint (2,207.1 MB, saving 505.6 MB over `FP16` and 166.0 MB over `INT8`) while sustaining 16.01 tok/s decoding throughput (2.77× faster than `LLM.int8()`) and 578.19 tok/s prefill speed (3.04× faster than `LLM.int8()`). Consequently, `FP16` and `NF4` define the deployment Pareto frontier on Turing GPUs.

Task Regime	Target Subspace	Metric 1	Metric 2	Overall	VRAM (MB)
Syntax-Constrained Regime: JSONOut (4 ep, r = 16, grad_accum = 4)					
JSONOut	qv (W_q, W_v)	Parse: 1.0000	Field: 1.0000	**1.0000**	**1935.9**
JSONOut	qkvo ($\mathcal{M}_{\text{attn}}$)	Parse: 1.0000	Field: 1.0000	**1.0000**	1948.3
JSONOut	all-linear ($\mathcal{M}_{\text{all}}$)	Parse: 1.0000	Field: 1.0000	**1.0000**	2024.3
Knowledge-Intensive Regime: CourseQA (3 ep, r = 16, grad_accum = 2)					
CourseQA	qv (W_q, W_v)	EM: 0.0000	F1: 0.2767	0.1887	**1933.3**
CourseQA	qkvo ($\mathcal{M}_{\text{attn}}$)	EM: 0.0625	F1: 0.3853	0.2717	1945.4
CourseQA	all-linear ($\mathcal{M}_{\text{all}}$)	**EM: 0.1250**	**F1: 0.4343**	**0.3223**	2021.3

Table 5: Target subspace ablation on `Qwen2.5-0.5B` contrasting attention-only (`qv`, `qkvo`) against full linear (`all-linear`: attention + SwiGLU MLP) adapters across syntactic (JSONOut) and factual (CourseQA) regimes.

4.3 Orthogonal Ablations and Subspace Allocation (Q1, Q2)

Tables 4 and 5 and Figure 3(a–c) report four orthogonal ablations holding all non-ablated hyperparameters constant:

Sample scaling and label-noise phase transition. In Table 4(A), $N = 25$ examples (∼ 3 per topic) fail to reach the critical sample threshold on CourseQA (0.0767 Overall); doubling to $N = 50$ yields a 3.49× jump to 0.2675, and $N = 104$ reaches 0.3680. In Table 4(B), `MiniMind-26M` exhibits strong resilience to moderate label corruption on Intent: at $\eta_{\text{noise}} = 20\%$, test accuracy remains 97.92% (Macro-F1 97.83%), as coherent clean-label gradients dominate early optimization (Arpit et al., 2017). At $\eta_{\text{noise}} = 50\%$, accuracy drops to 56.77% (Macro-F1 47.43%), yet the unmapped rate remains 0.00%—indicating that output vocabulary syntax is preserved even under severe semantic label noise.

The inverted-U LoRA rank curve. Sweeping $r \in \{4, 8, 16, 32\}$ with $\alpha = 2r$ on `Qwen2.5-0.5B` (CourseQA, Table 4C, Figure 3b) reveals a non-monotonic curve peaking at $r = 4$ (Overall 56.88%, EM 50.00%, Token-F1 69.95%, Contains 53.12%) with the lowest memory footprint (1,948.1 MB). Increasing rank to $r = 32$ lowers Overall by −5.40% to 51.48% (Token-F1 62.43%) while consuming +174.8 MB additional VRAM. This confirms Proposition 2: on compact domain datasets, minimal rank $r = 4$ captures the task subspace while higher ranks reintroduce overfitting.

Subspace dichotomy between syntactic routing and factual memory. Table 5 and Figure 3(c) test Inequality (5) directly on `Qwen2.5-0.5B`. On JSONOut, adapting $\mathcal{M}_{\text{qv}} = \{W_q, W_v\}$ alone achieves 100.0% schema validity and 100.0% field accuracy; expanding to `qkvo` or `all-linear` provides zero accuracy lift while adding 88.4 MB of VRAM overhead. On CourseQA (evaluated at 3 epochs, grad_accum = 2 to isolate early subspace learning speed), `qv`-only adaptation reaches only 18.87% Overall (EM 0.00%, Token-F1 27.67%, Contains 3.12%). Unlocking full attention (`qkvo`) lifts Overall by +8.30% to 27.17% (EM 6.25%, Token-F1 38.53%), and unlocking the SwiGLU FFN projections (`all-linear`: `gate_proj`, `up_proj`,

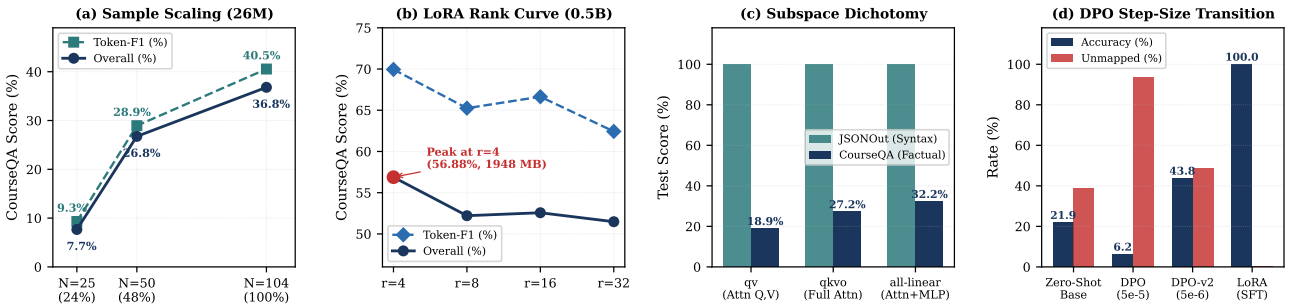


Figure 3: Orthogonal ablations and preference alignment diagnostics. (a) Sample-size scaling on `MiniMind-26M` (CourseQA, 8 ep Full SFT). **(b)** Inverted-U LoRA rank curve on `Qwen2.5-0.5B` (CourseQA, 6 ep), peaking at $r = 4$ (56.88%, Token-F1 69.95%). **(c)** Subspace allocation dichotomy: `qv` saturates syntactic JSONOut (100.0%), whereas factual CourseQA climbs monotonically from `qv` (18.87%) to `qkvo` (27.17%) and `all-linear` (32.23%). **(d)** DPO learning-rate phase transition on `Qwen2.5-0.5B` (Intent), where $\eta = 5 \times 10^{-6}$ recovers from likelihood displacement collapse.

Alignment Regime	Hyperparameters	Acc	Macro-F1	Unmapped	VRAM (MB)
Zero-Shot Baseline	—	0.2188	0.1903	0.3854	2024.8
DPO (High Step)	$\beta = 0.1, \eta = 5 \times 10^{-5}$	0.0625	0.0889	0.9375	2025.4
DPO (Bounded Step)	$\beta = 0.1, \eta = 5 \times 10^{-6}$	**0.4375**	**0.4759**	0.4844	**971.2**
Supervised LoRA	$r = 16, \eta = 3 \times 10^{-4}$	**1.0000**	**1.0000**	**0.0000**	2025.4

Table 6: Preference alignment stability on `Qwen2.5-0.5B` (Intent, $n = 192$ test queries) across DPO step sizes vs. Zero-Shot and Supervised LoRA.

`down_proj`) adds a further +5.06% to reach 32.23% (EM 12.50%, Token-F1 43.43%, a +13.36% total gain over `qv`). This confirms that factual adaptation requires modifying FFN key-value memories (Geva et al., 2021), whereas syntactic formatting only requires attention re-routing.

4.4 Preference Alignment: DPO Step-Size Phase Transition (Q4)

Table 6 and Figure 3(d) evaluate DPO on `Qwen2.5-0.5B` (Intent), where each training prompt x pairs its ground-truth label y_w against a corrupted negative label y_l.

At $\eta = 5 \times 10^{-5}$, DPO exhibits severe likelihood displacement (Section 2.4): accuracy drops from 21.88% (Zero-Shot) to 6.25%, while the unmapped rate spikes to 93.75% as the policy emits off-manifold English phrases (e.g., "good weather!") that share zero tokens with the penalized snake_case labels. Lowering the step size by one order of magnitude to $\eta = 5 \times 10^{-6}$ (`dpo-v2`) stabilizes the reference KL constraint, lifting accuracy by 7.0× to 43.75% and Macro-F1 to 47.59% (2.0× above Zero-Shot). Even in the stable regime, however, standalone DPO trails supervised LoRA (100.0%), because pairwise repulsion against a single negative label y_l per query does not suppress the remaining four incorrect classes.

Claim boundary. Propositions 1 and 2 characterize hardware latency bounds and linear update rank ceilings; they do not claim universal numerical values for r^* across arbitrary pre-training corpora. Empirically, our 100.0% results on JSONOut and Intent establish that 26M–0.5B SLMs can master strict 4-schema extraction and 6-way out-of-template classification; they do not imply that 26M models match frontier LLMs on open-ended reasoning, where capacity bounds remain binding (CourseQA 45.71% at 26M vs. 58.80% at 1.5B).

5 Related Work

Parameter-efficient adaptation and intrinsic dimensionality. Aghajanyan et al. (2021) showed that pre-trained language models possess a low intrinsic dimension during fine-tuning, motivating LoRA (Hu et al., 2022), adaptive rank allocation (Zhang et al., 2023), and quantized backpropagation via QLoRA (Dettmers et al., 2023). While prior work primarily treats LoRA as a memory-reduction approximation for 7B+ models, we show that in sub-billion SLMs (26M–0.5B) trained on compact domain datasets, rank truncation ($r \in [4, 16]$) acts as a spectral regularizer that substantially outperforms full-parameter SFT.

Mechanistic localization and edge quantization. Geva et al. (2021) and Meng et al. (2022) demonstrated that Transformer MLP layers operate as key-value memories encoding factual knowledge. Our target-subspace ablation connects this mechanistic finding to PEFT design, explaining why `qv` adapters saturate syntactic extraction while MLP adapters are required for domain QA. On the deployment side, `LLM.int8()` (Dettmers et al., 2022) and `NF4` (Dettmers et al., 2023) compress model weights, but our Roofline analysis (Williams et al., 2009) and Turing `sm_75` benchmarks reveal that dequantization kernel latency governs batch-1 decoding throughput.

Direct preference optimization. Rafailov et al. (2023) formulated DPO as a closed-form alternative to RLHF. Subsequent studies identified likelihood displacement under offline preference training (Pal et al., 2024; Tajwar et al., 2024); our results isolate the step-size phase transition governing this failure mode on discrete classification manifolds.

6 Limitations

Hardware scope and batching regime. Our Roofline latency measurements are conducted on an NVIDIA Turing `sm_75` GPU at batch size $B = 1$ to reflect single-user edge deployment. On newer Hopper/Blackwell architectures with native `FP8`/`INT4` tensor instructions or under large-batch continuous batching ($B \geq 32$) where decoding becomes compute-bound ($\mathcal{I}_{\text{decode}} > \mathcal{I}^*$), the relative throughput ratio between `INT8` and `FP16` will shift.

Capacity ceiling on multi-hop reasoning. Qualitative error inspection on CourseQA shows that while `MiniMind-26M` LoRA masters single-concept definitions, it still confuses closely related multi-hop concepts (e.g., conflating Prim's and Kruskal's greedy strategies on minimum spanning trees), whereas `Qwen2.5-1.5B` QLoRA resolves them. Low-rank adaptation optimizes extraction of pre-trained representations but cannot substitute for base-model capacity when pre-training exposure is absent.

7 Conclusion

We formulated and empirically validated a first-principles framework for adapting and quantizing Small Language Models (26M–1.5B) under edge hardware constraints. Across 39 controlled runs, we showed that (i) low-rank adaptation ($r \in [4, 16]$) acts as a spectral regularizer that outperforms Full SFT by +8.91% on compact domain QA; (ii) attention-only (`qv`) adapters saturate syntactic JSON extraction (100.0%), whereas domain QA requires unlocking FFN projections (+13.36% gain); (iii) 4-bit `NF4` dominates `LLM.int8()` by 2.77× in batch-1 decoding throughput while saving 505.6 MB of static VRAM; and (iv) bounding DPO step sizes to $\eta \leq 5 \times 10^{-6}$ prevents likelihood displacement on compact classification manifolds.

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